

Title Page

- Lab Title: USB Image Acquisition and Hash Verification
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- Course Name: Computer And Forensics
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- Date of Submission: 28/8/2026

Executive Summary

This report documents the forensic acquisition of a USB flash drive. The acquisition followed industry-standard forensic procedures using FTK Imager and dd to create a bit-for-bit copy of the source media. Cryptographic hash verification (MD5 and SHA-256) confirmed the integrity of the acquired image. The image was validated to contain the required evidence file (hauwa_yahaya.txt), demonstrating successful acquisition and verification.

Lab Objectives

The objective of this lab is to demonstrate proficiency in forensic disk imaging procedures, acquire a forensically sound image of a test USB drive, verify image integrity using cryptographic hashing, validate the contents of the acquired image and document the complete acquisition process

Tools and Resources Used

Methodology

Part A - Prepare the USB Evidence

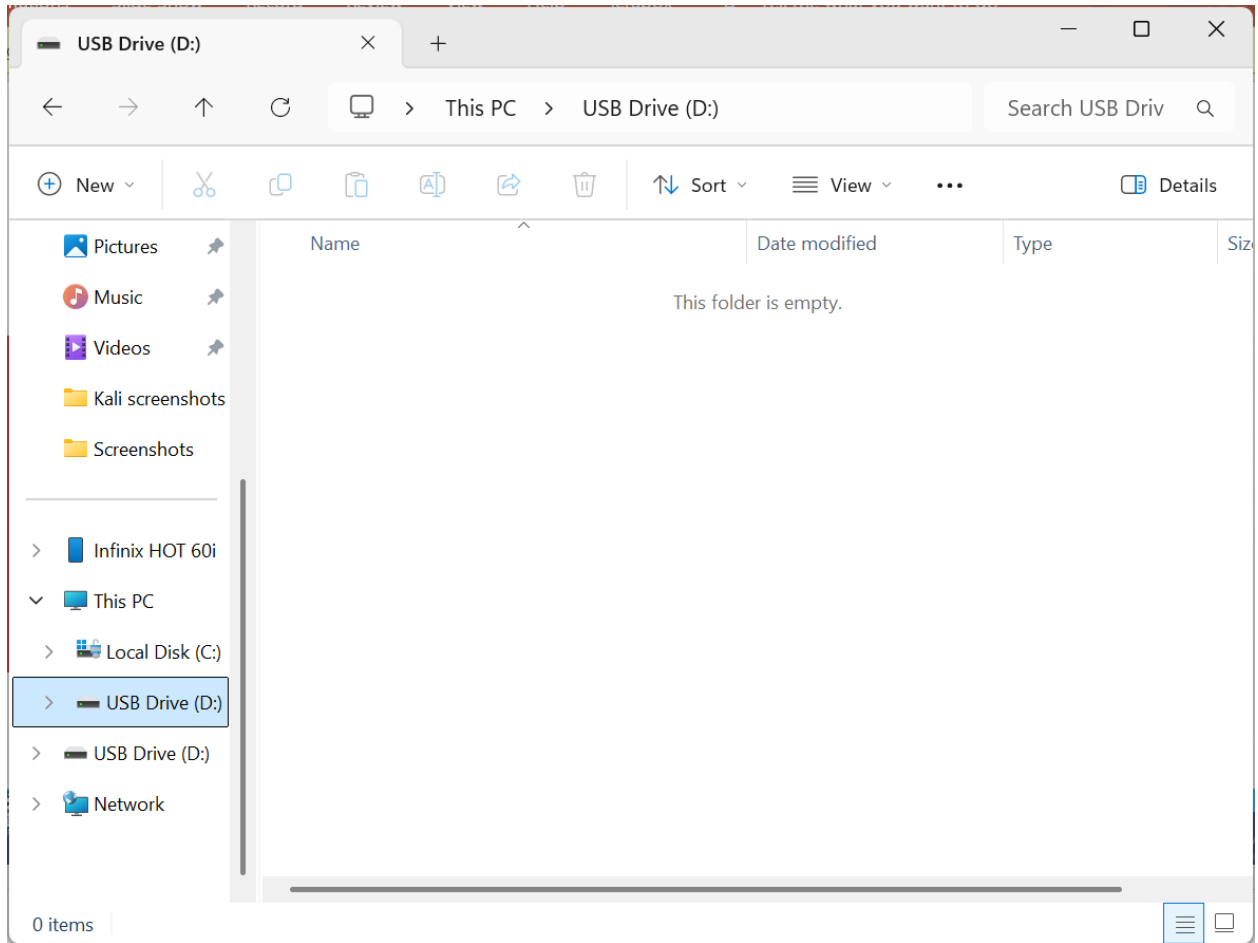


FIGURE 1: Windows explorer showing plugged USB drive D

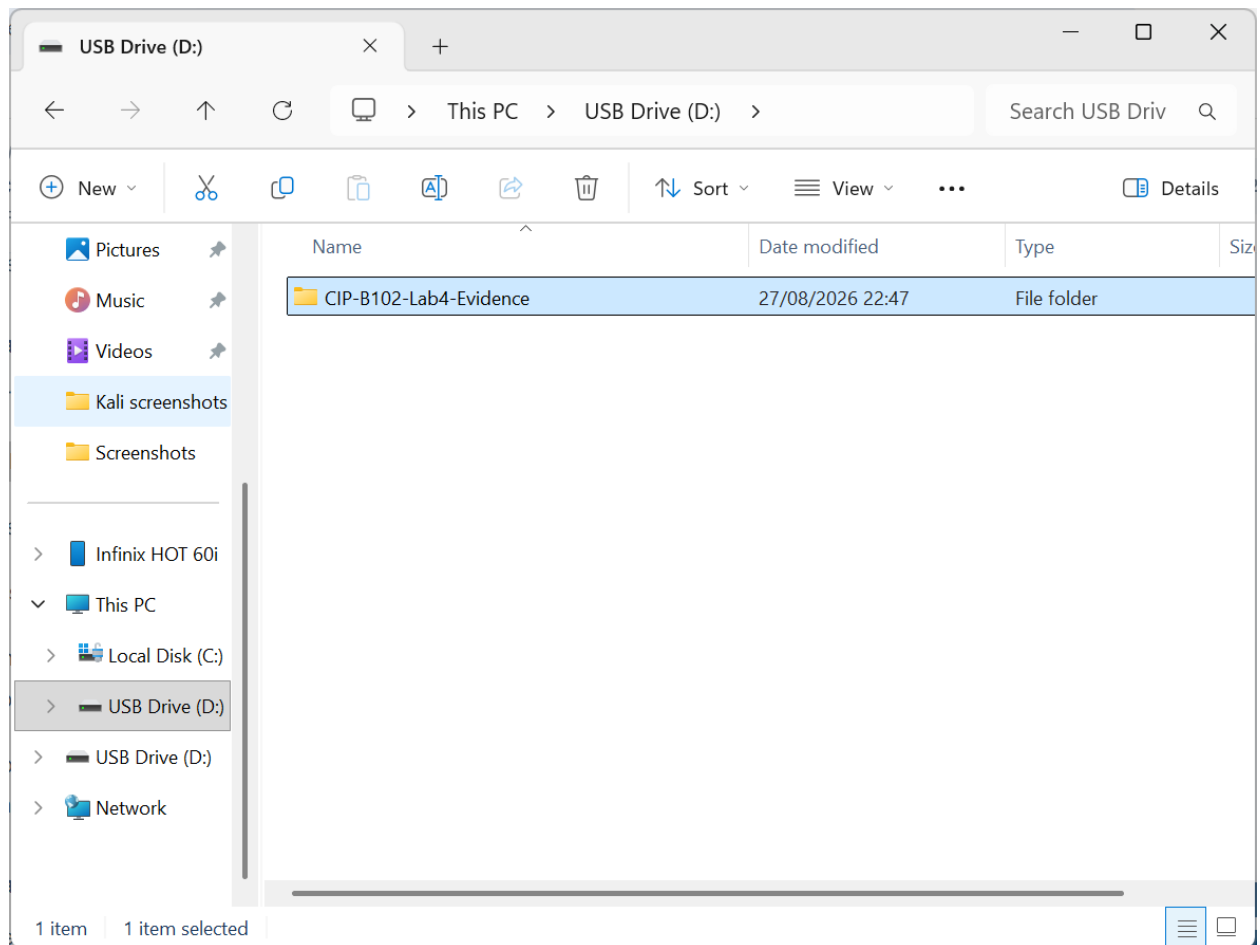


FIGURE 2: Screenshot of the newly created folder

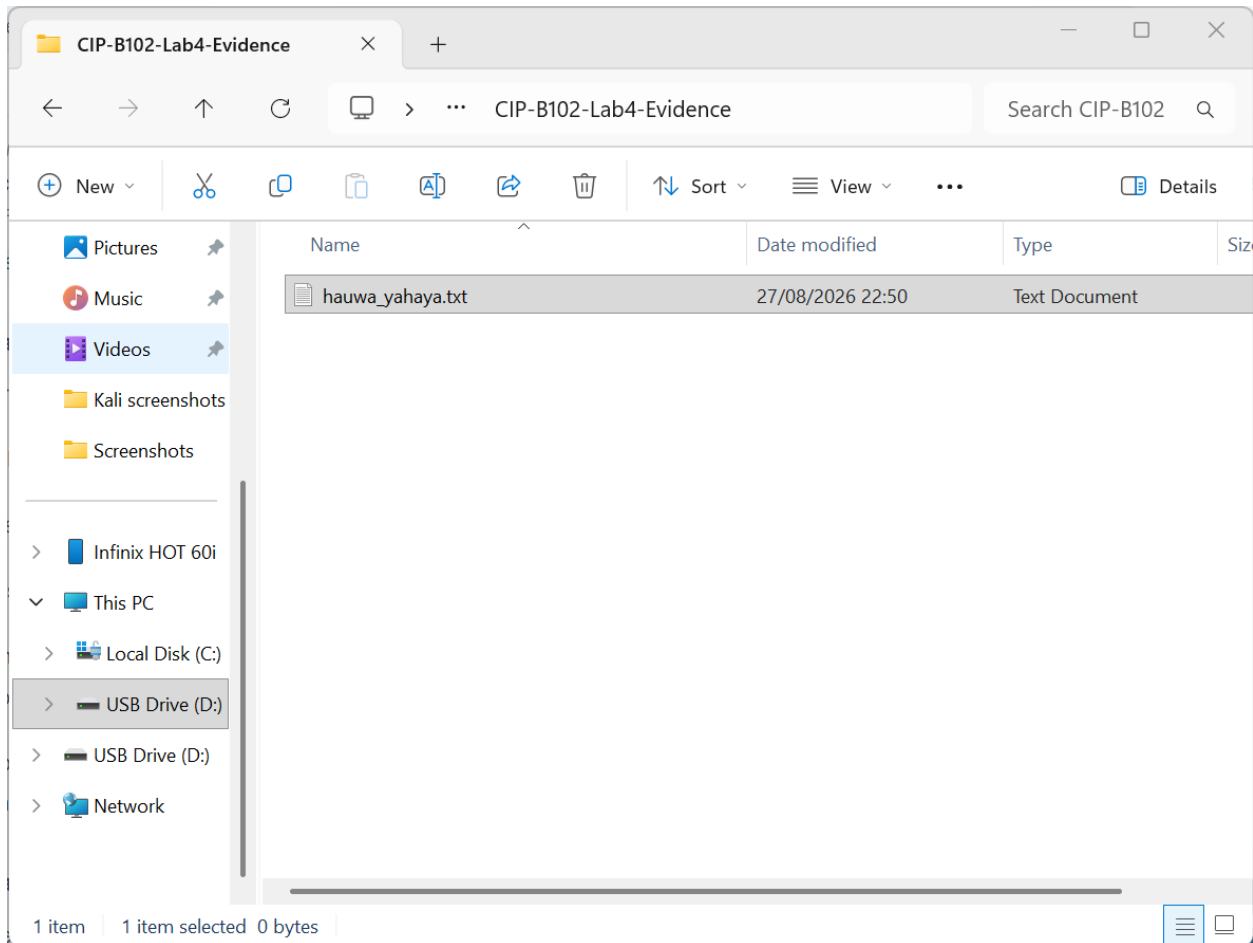


FIGURE 3: Screenshot of the newly created text file

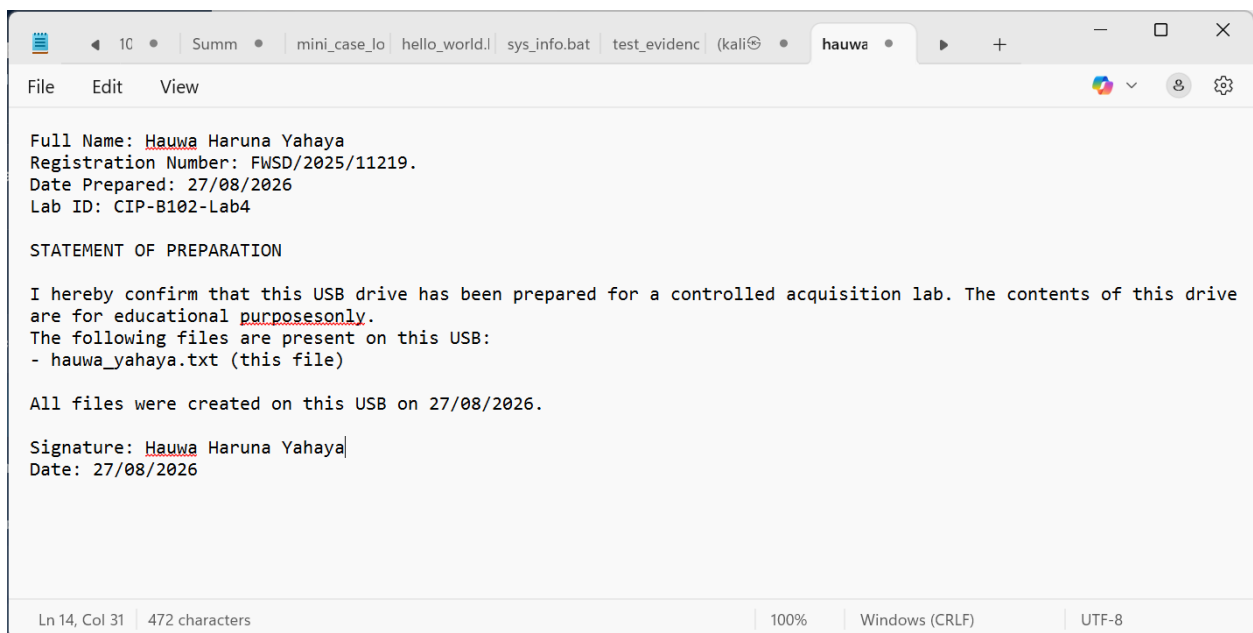


FIGURE 4: Contents of the text file.

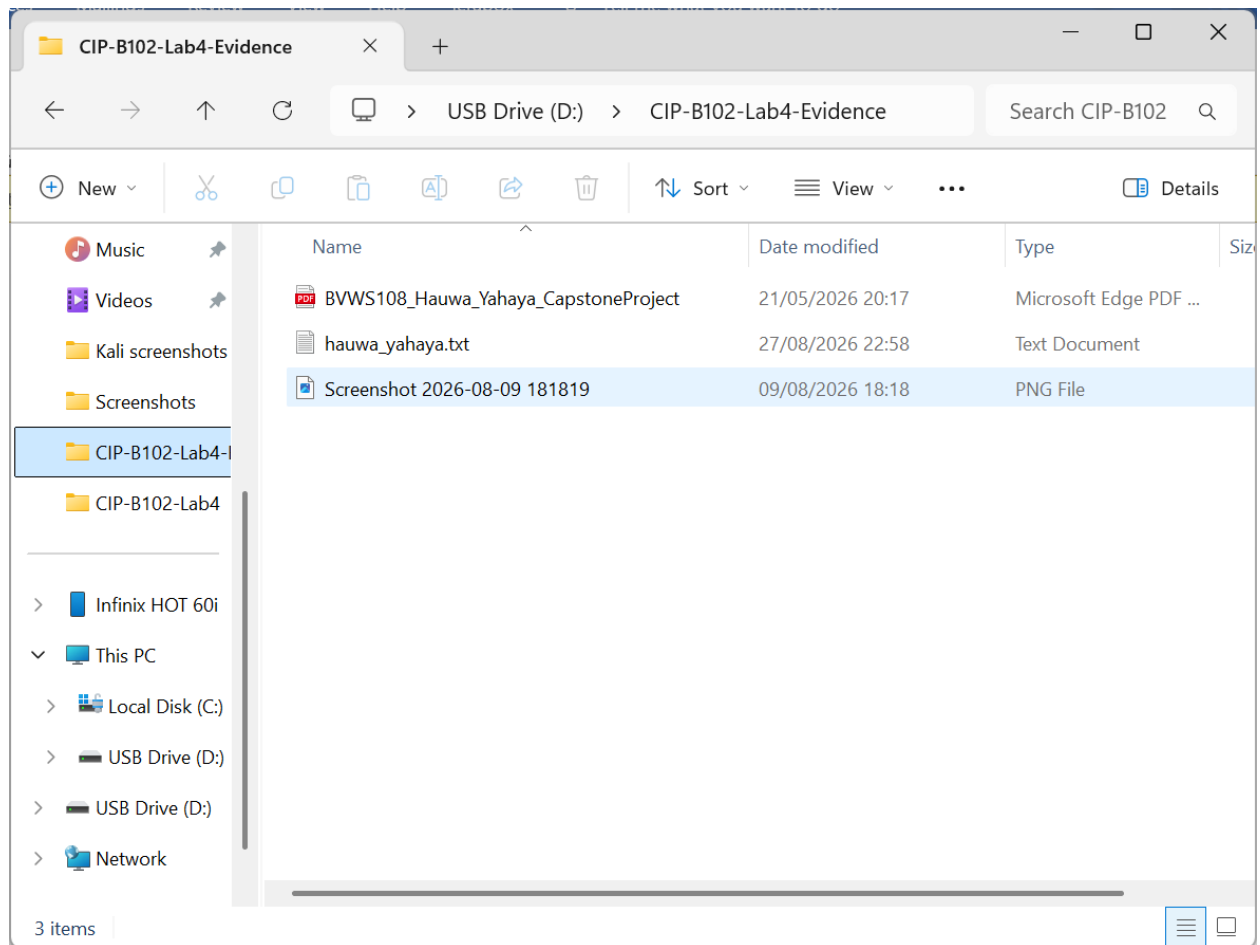


FIGURE 5: I added one image and one pdf file to the parent folder

Part B - Complete Evidence Documentation

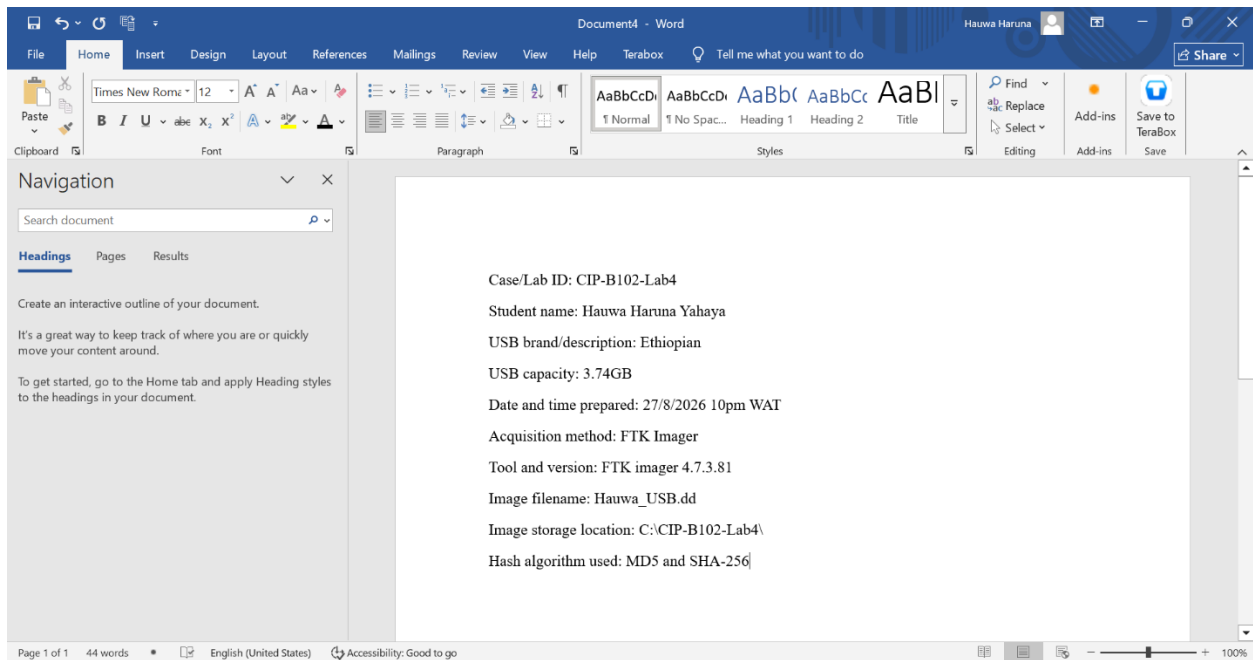


FIGURE 6: Screenshot of evidence documentation

Part C - Acquire the USB Image

Option A - Windows using FTK Imager

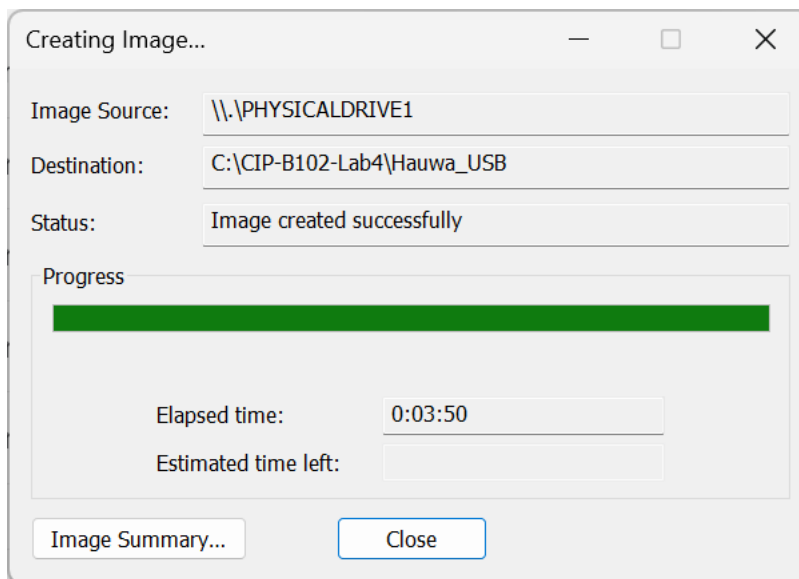


FIGURE 7: FTK Imager showing the acquisition progress bar

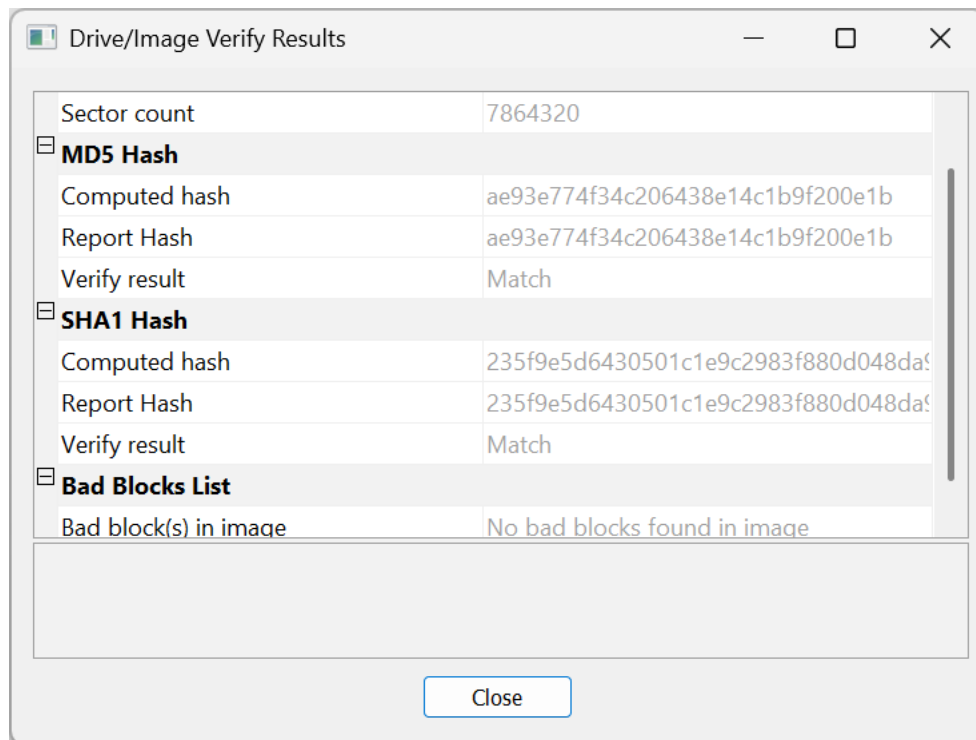


FIGURE 8: FTK Imager showing "Image Verify Result: MATCH" with the hash values

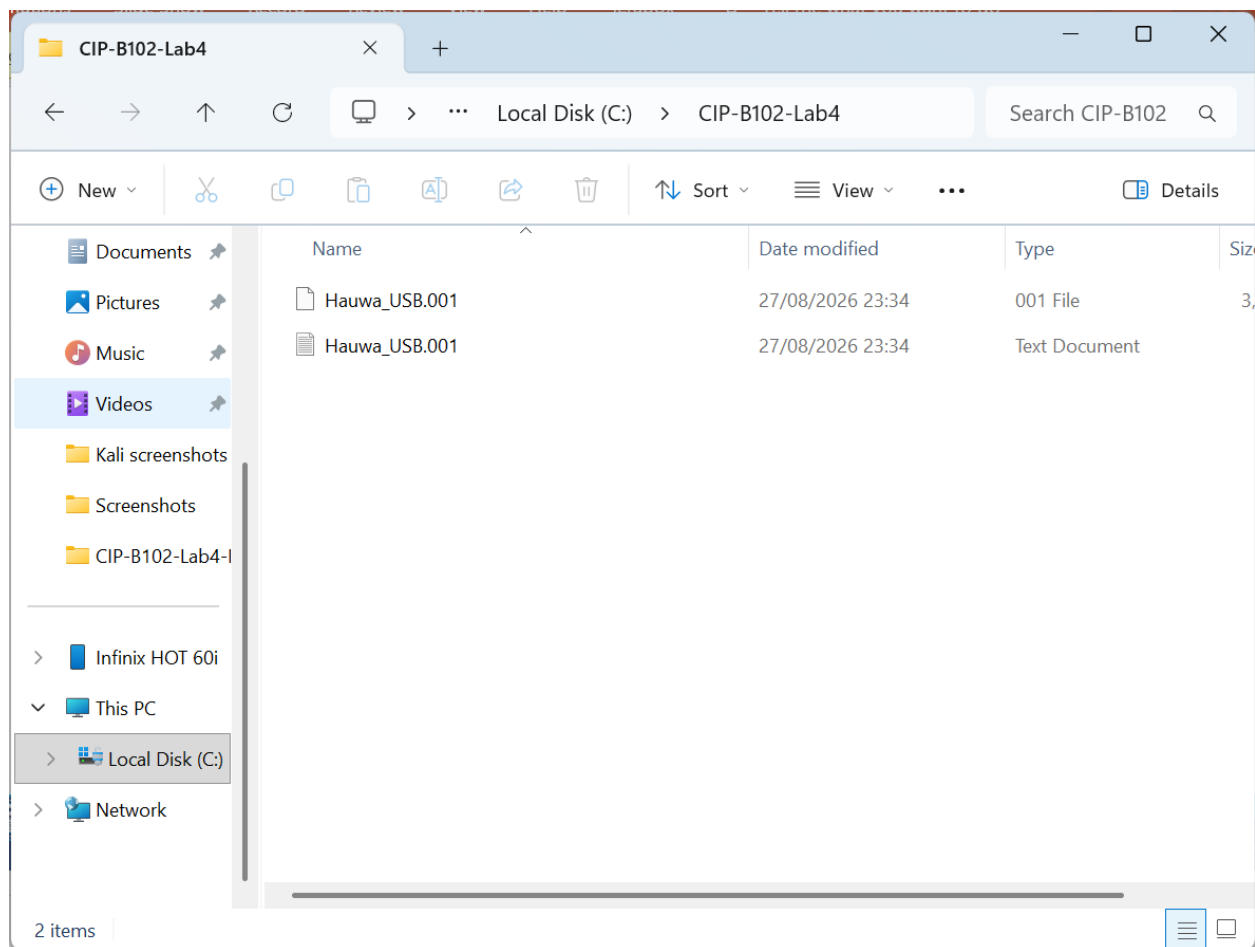


FIGURE 9: Screenshot of the disk image on my computer.

Part E - Validate the Acquired Image

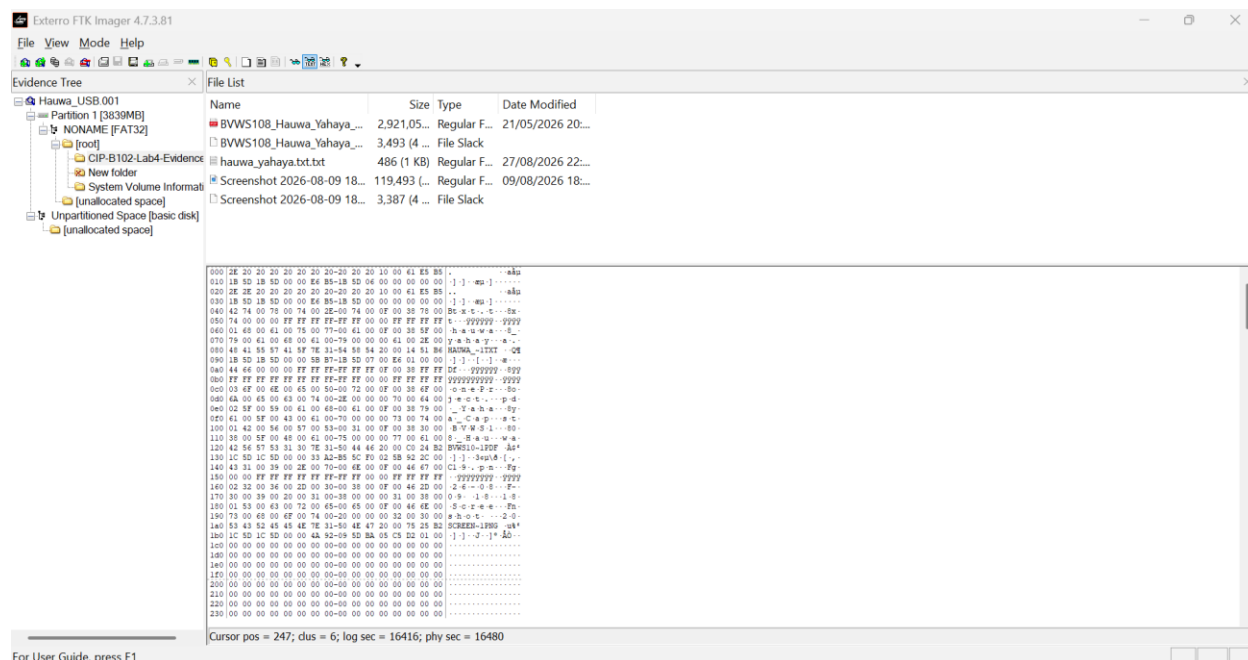


FIGURE 10: I confirm that the acquired image contains hauwa_yahaya.txt

REFLECTION

This laboratory exercise provided valuable hands-on experience in forensic USB acquisition using FTK Imager on Windows. The process was generally smooth, with the evidence preparation and documentation phases being straightforward. The most critical learning point was the importance of careful device identification, as selecting the wrong drive could have resulted in catastrophic data loss. I initially encountered confusion when FTK Imager saved the image as a .001 file rather than a .dd file, but I quickly learned that this is the standard segmentation format used by the tool. The automatic hash verification feature in FTK Imager was extremely helpful, eliminating the need for manual hash calculations and reducing the risk of errors. Overall, this lab reinforced the fundamental principles of forensic imaging, particularly the importance of hash verification for ensuring evidence integrity. In future acquisitions, I would ensure that I take more screenshots during intermediate stages and verify the image contents immediately after acquisition rather than waiting until the end.

CONCLUSION

This laboratory exercise successfully demonstrated the complete forensic acquisition process for a USB flash drive using FTK Imager on Windows. The objectives were fully met, including evidence preparation, forensic imaging, hash verification and image validation. The MD5 and SHA-256 hash values of the source USB and the acquired image matched identically, confirming the integrity of the forensic image. When the image was mounted and browsed in FTK Imager, the hauwa_yahaya.txt file was successfully located and its contents verified against the original. This confirms that the acquisition was forensically sound and that the image is an exact bit-for-

bit copy of the source device. The experience has reinforced the importance of meticulous documentation, careful device selection and hash verification in digital forensic investigations. I am now confident in my ability to perform forensic USB acquisitions and produce professional acquisition reports in accordance with industry standards.